

INDU 211/2X 2025 – Assignment #4

Problem 1 (Chapter 14, Problem #8)

8. Solve graphically for the values of x_1 and x_2 that

subject to:

$$\begin{aligned}\text{minimize } TC &= 7x_1 + 9x_2 \\ 18x_1 + 7x_2 &\geq 250 \\ 3x_1 + 5x_2 &\geq 84 \\ 9x_1 + 2x_2 &\geq 60 \\ x_1 &\geq 0 \quad x_2 \geq 0\end{aligned}$$

Problem 2 (Chapter 14, Problem #10)

10. A small machine shop has capacity to do turning, milling, drilling, and welding. Two products are under consideration. Each will yield a net profit of \$3.50 per unit and will require the following machine time:

Operation	Product		Capacity (hours)
	1	2	
Turning	0.092	0.248	32
Milling	0.224	0.096	16
Drilling	0.080	0.000	32
Welding	0.000	0.240	16

- (a) Formulate this as a linear programming problem similar to Equations (14.38) to (14.42).
- (b) Solve graphically for the number of units of each product that should be scheduled to maximize profit.